

LD4B-780-FP-100

OVERVIEW

Laser diode coupled to an optical fiber and packaged into a hermetic case.

MAIN FEATURES

- Wavelength: 780 nm
- Cavity type: Fabry-Perot
- Optical power in CW mode in single-mode fiber: 100 mW
- Package types: coaxial, coaxial with bracket, 14 pins DIL, 14 pins BTF

ORDERING INFORMATION

LD4B-780-FP-100-X-X-X-X-X-X

Case type

COAX: compact coaxial (low duty cycle pulse mode only)

COAXB: compact coaxial with a bracket

TH: compact coaxial with a bracket compatible to Thorlabs mount

DIL: common 14-pins DIL for active thermal stabilization (TEC and thermistor)

DILRAD: 14-pins DIL for active thermal stabilization (TEC and thermistor) with wall radiator

BTF: 14-pins BTF type 1 (Pump) for active thermal stabilization (TEC and thermistor)

Pinout code

18: see more details on page 5

Fiber type

SM05: SM, [Corning HI-780](#), furcation tubing Ø0.9 mm

SMP05: PM, [Coherent PM780-HP](#), furcation tubing Ø0.9 mm

MM5: MM, [50/125, OM3](#), furcation tubing Ø0.9 mm

MM6: MM, [62.5/125, OM1](#), furcation tubing Ø0.9 mm

Other type on request

Connector type

FU: FC/UPC (SM05, SMP05)

FA: FC/APC (SM05, SMP05)

N: no connector (scissors cut)

Other type: on request

Test measurements

CW: CW mode (electro-optical parameters at T=25+/-5 C and spectrum)

P: Pulse mode (pulse duration 5 us, duty cycle 1%, at T=25+/-5 C)

CWP: both CW and pulse mode

Fiber length

0.5: 500+/-50 mm

1.0: 1000+/-100 mm

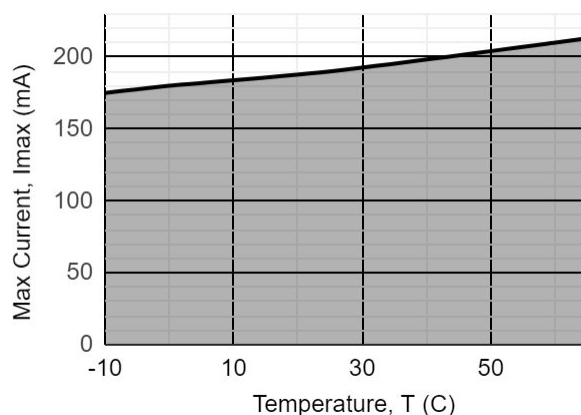
Other length on request

LD4B-780-FP-100

ABSOLUTE MAXIMUM RATINGS

Parameter		Value	Unit	Conditions
Laser diode forward current*	I_{max}	210	mA	CW, $T = 25^{\circ}\text{C}$
Laser diode reverse voltage	V_{RL}	2	V	
Operating temperature**	T_{OP}	-10 - +65	$^{\circ}\text{C}$	Coaxial package
Operating temperature**	T_{OP}	-40 - +60	$^{\circ}\text{C}$	DIL, BTF ($T_{st} = 25^{\circ}\text{C}$)
Storage temperature	T_{stg}	-40 - +85	$^{\circ}\text{C}$	
Soldering temperature	T_{sold}	260	$^{\circ}\text{C}$	Max. 5 seconds

*Maximal laser diode forward current depends on the operating temperature. Please, refer to the figure below.



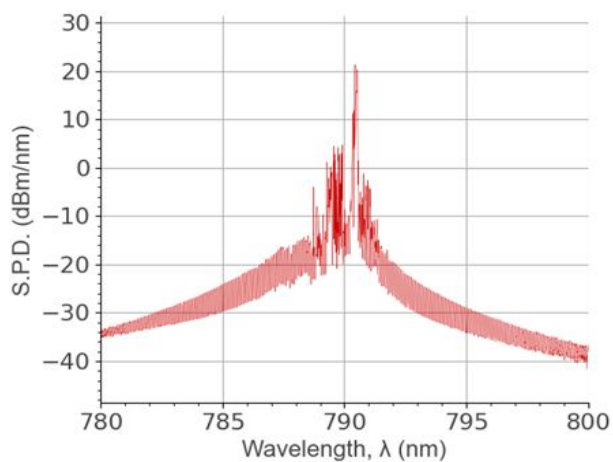
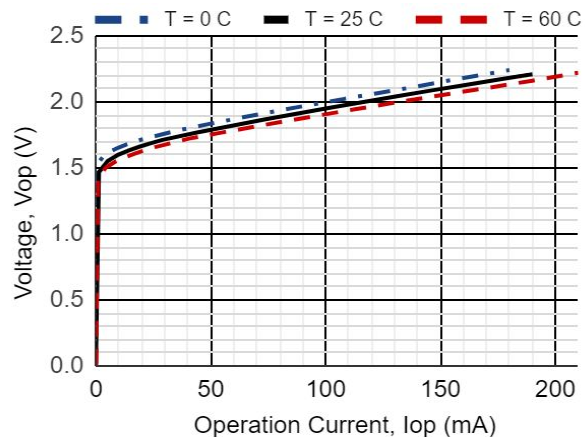
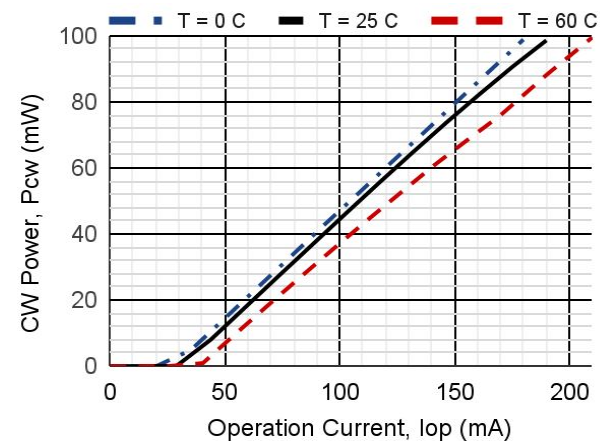
**Operating temperature is defined by the case temperature. It is recommended to ensure sufficient heat dissipation so that the module's maximum operating temperature is not exceeded.

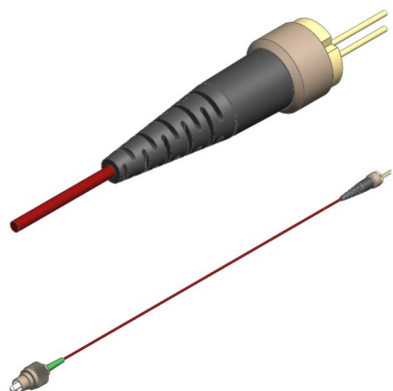
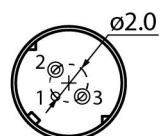
Operating temperature for the DIL, DILRAD and BTF 14-pins case T with TEC is defined for internal temperature stabilization at $T_{st} = 25^{\circ}\text{C}$ that corresponds to thermistor resistance $R_t = 10 \text{ k}\Omega$.

LD4B-780-FP-100

ELECTRICAL-OPTICAL CHARACTERISTICS (T = 25 °C)

Parameter		MIN	TYP	MAX	Unit	Conditions
Optical power (CW)	P _{cw}	90	100		mW	CW, I _{op} = 190 mA, SM05
Mean wavelength	λ	770	780	795	nm	CW, I _{op} = 190 mA
Spectral width	Δλ		1	3	nm	CW, I _{op} = 190 mA
Wavelength-temperature coefficient	dλ/dT		0.27		nm/°C	CW, I _{op} = 190 mA
Threshold current	I _{th}		30	40	mA	
Slope efficiency	S _e	0.55	0.62		mW/mA	CW, SM05
Operating voltage	V _{op}		2.2	2.5	V	CW, I _{op} = 190 mA
Polarization extinction ratio	PER	17				CW, I _{op} = 190 mA, SMP05

LD4B-780-FP-100

LD4B-780-FP-100**COAX****BACK VIEW****PINOUT**

#18

2_oDownload more
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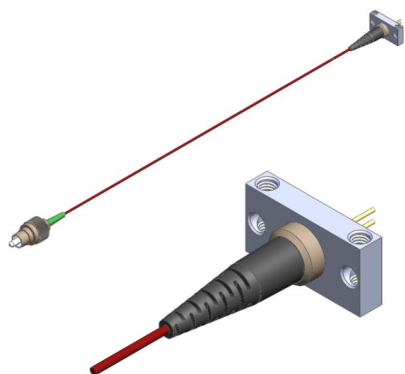
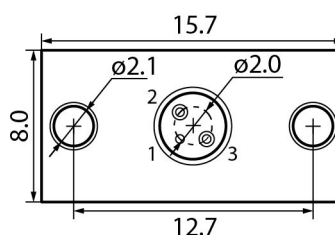
Drawing



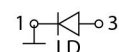
3D model



Application Notes

**COAXB****BACK VIEW****PINOUT**

#18

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information

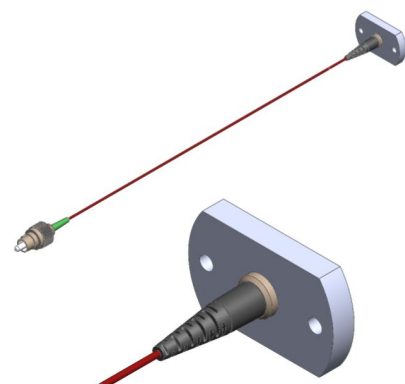
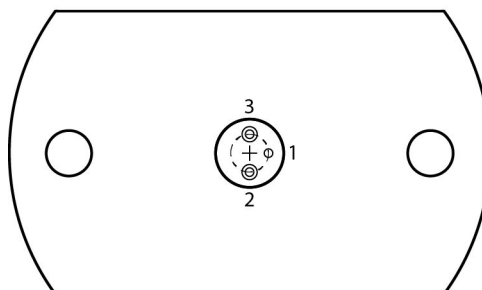
Drawing



3D model

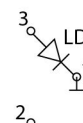


Application Notes

**PACKAGE TH****BACK VIEW****PINOUT**

#18

Thorlabs Pin Code H

Compatible to Thorlabs
LDM9LP mountDownload more
information

Drawing



3D model

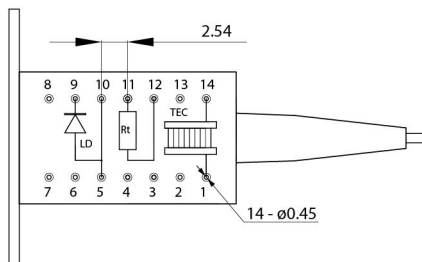


Application Notes

LD4B-780-FP-100**DIL**

PINOUT #4, #18, #21, #31

BOTTOM VIEW



- 1.TEC Anode
- 2.-
- 3.-
- 4.-
- 5.LD Anode
- 6.-
- 7.-
- 8.-
- 9.LD Cathode
- 10.LD Anode
- 11.Thermistor
- 12.Thermistor
- 13.-
- 14.TEC Cathode

TEC HP: $I_{\max} = 1.4A$, $U_{\max} = 3.9V$, $Q_{\max} = 3.3W$,
 $AC R = 2.0 \Omega$, $\Delta T_{\max} = 69 K$

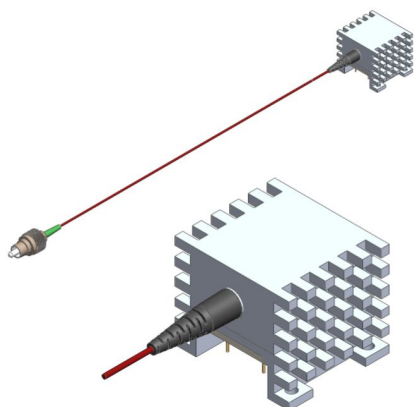
Thermistor:
 $R_t = 10 \cdot \exp(3600 \cdot \{1/T[K] - 1/298\}) \text{ k}\Omega$

Download more information

Drawing

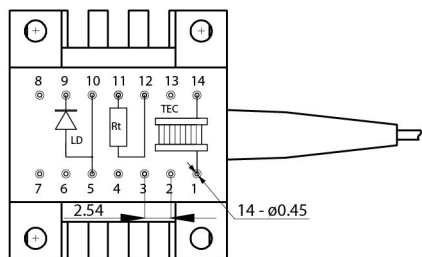
3D model

Application Notes

**DILRAD**

PINOUT #4, #18, #21, #31

BOTTOM VIEW



- 1.TEC Anode
- 2.-
- 3.-
- 4.-
- 5.LD Anode
- 6.-
- 7.-
- 8.-
- 9.LD Cathode
- 10.LD Anode
- 11.Thermistor
- 12.Thermistor
- 13.-
- 14.TEC Cathode

TEC HP: $I_{\max} = 1.4A$, $U_{\max} = 3.9V$, $Q_{\max} = 3.3W$,
 $AC R = 2.0 \Omega$, $\Delta T_{\max} = 69 K$

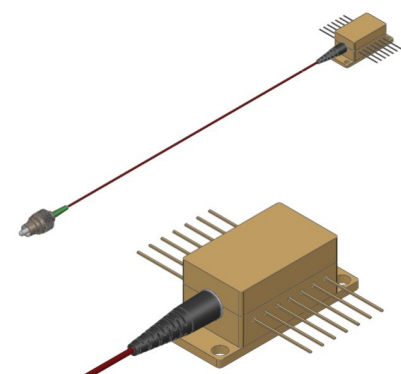
Thermistor:
 $R_t = 10 \cdot \exp(3600 \cdot \{1/T[K] - 1/298\}) \text{ k}\Omega$

Download more information

Drawing

3D model

Application Notes

**BTF**

- 1. TEC+
- 2. Thermistor
- 3. NC
- 4. NC
- 5. Thermistor
- 6. NC
- 7. NC
- 8. NC
- 9. NC
- 10 LD/SLD Anode
- 11. LD/SLD Cathode
- 12. NC
- 13. Case Ground
- 14. TEC -

TEC HP: $I_{\max} = 1.4A$, $U_{\max} = 3.9V$, $Q_{\max} = 3.3W$,
 $AC R = 2.0 \Omega$, $\Delta T_{\max} = 69 K$

Package limits RF bandwidth to 500 MHz
 Thermistor:
 $R_t = 10 \cdot \exp(3600 \cdot \{1/T[K] - 1/298\}) \text{ k}\Omega$

Download more information

Drawing

3D model

Application Notes

LD4B-780-FP-100

Characteristics, data, materials and structures specified in this datasheet are subject to change without notice. Please refer to the latest specification before use of the products.

Safety and handling cautions

1. Avoid smashing and burning of the module. Avoid storing and using the module in conditions where water, organic solvents or aggressive acids or bases may contact the module or where there is a possibility of exposure to corrosive gases, explosive gases, dust, salinity or other harsh conditions. The module should be disposed as special industrial waste.
2. Exceeding absolute maximum ratings even for a short time can cause permanent damage of the module.
3. The module is sensitive to and can be broken by ESD (static electricity).

Conflict Minerals Policy Statement

LD4B achieves business objectives and customer needs with social responsibility. We do not support or contribute to the violence and human rights violations associated with the mining of conflict minerals coming from Conflict Regions according to US "Dodd-Frank Act". When possible, our suppliers' conflict mineral statements are reviewed. We do not directly purchase Conflict Minerals from any source and do not knowingly procure any parts and products containing Conflict Minerals from Conflict Regions.

RoHS Compliance Statement

Restriction of Hazardous Substances (RoHS) directive (Directive 2011/65/EC amended with Directive (EU) 2015/863) is the directive aimed at reducing the harmful environmental impact of waste electrical equipment by restricting the use of known dangerous substances. Based on information received from our supply sources, LD4B hereby states that the banned substances listed in the RoHS directive are not found in the parts and materials used above the threshold level listed other than exceptions approved by the European Commission.

REACH Compliance Statement

Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) is a European Union regulation 1907/2006/EC that addresses the production and use of chemical substances, and their potential impacts on human health and the environment. Based on information received from our supply sources, LD4B hereby states compliance of the parts and materials used in manufacturing to REACH regulation. LD4B does not manufacture or import any substances or preparations as defined under REACH.